CCTE R82 Lab Scripts: lab menus and improvements

Handout for instructors and students · scripts 2.2.5, A-GUI menu 1.0.4 · 8 November 2026 · Don Paterson

Each CCTE lab starts with a **lab script** that creates the problem you troubleshoot. In this class the scripts have been updated and two optional menus added. **Nothing in the lab guide changes:** every command in the book still works exactly as written and sets up the same lab. The menus are simply easier ways to run the same commands.

1. Three ways to run a lab script

	As in the lab guide	labmenu (on the appliance)	CCTE Lab Menu (on A-GUI)
Where	SSH to A-SMS (A-GW for Lab 4A), Expert mode	Same, then type `labmenu`	Desktop shortcut on A-GUI; no SSH needed
Start	`./labstart lab8a` (in /home/admin/CCTE_Scripts)	choose the lab, **1) Start**	choose the lab, **1) Start**
Stop / Revert	`./labstop lab8a` `./labrevert lab8a`	**2) Stop** / **3) Revert**	**2) Stop** / **3) Revert**
Lab 2B	Two SSH windows; follow the prompts	Run it on A-SMS and A-SMS-02 in turn	**One window**: drives both servers in the book's order
Policy installs	Check Recent Tasks in SmartConsole	Waits and shows each result	Waits and shows each result
Status	-	**4) Status** (read-only)	**4) Status** (read-only)

Both menus print the exact command they run, for example `Running: ./labstart lab8a`, so what you see always matches the book.

2. labmenu on the appliances

Log in to A-SMS as admin, type `expert` and the password, then `labmenu`. Choose a lab number, then an action. `v` turns verbose output on and off, `b` goes back, `q` quits. Lab 4A is on the gateways: run `labmenu` on the Active cluster member.

```
CCTE R82 Lab Scripts   host: A-SMS   scripts: 2.2.5
Labs on this host:
   1) LAB2A  Troubleshoot Security Management Server Processes  [not running]
   2) LAB2B  Troubleshoot Management High Availability
   3) LAB3A  Troubleshoot SmartLog  [not running]
   4) LAB5A  Debug the Security Gateway Kernel
   5) LAB6A  Debug Unified Policy Inspection
   6) LAB8A  Debug Identity Awareness
   7) LAB9A  Troubleshoot Site-to-Site VPN
  v) Verbose on/off   h) Help   q) Quit
```

3. CCTE Lab Menu on A-GUI

Double-click **CCTE Lab Menu** on the A-GUI desktop. Enter the lab admin password once; it is kept only while the window is open. Each action shows `Connecting to A-SMS (10.1.1.101) ... ok` and then the script output live. Each SSH login takes about 10 seconds.

Lab	What the A-GUI menu does for you
2B Start	Starts A-SMS (book 2B-1.2a), then A-SMS-02 (2B-1.2b), answers **y** (2B-1.2c) and types **exit** on A-SMS (2B-1.3). About one minute; ends with "Continue with Task 2B-2".
2B Stop	Book Task 2B-3: A-SMS, then A-SMS-02, and shows the `cpprod_util FwIsActiveManagement` checks. If both servers are still Active, it sets A-SMS-02 to Standby first and waits until A-SMS sees it (up to 5 minutes).
4A	Start runs on the Active cluster member; Stop finds the gateway where the lab is running and reminds you of step 4A-4.2 (start CPD again). Status shows both gateways.
5A, 6A, 8A, 9A	Waits for any policy installation still running, runs the script, then shows each installation result with the reason for any failure.

4. What the messages mean

You see	Meaning	What to do
`Setup completed. Kindly check policy installation status in SmartConsole`	The script did its job.	Carry on with the lab. In some labs (for example 8A) a policy install is **meant** to fail: that is the problem to troubleshoot.
`Lab lab8a is already set up (...). Nothing was changed.`	**New.** An earlier run already set the lab up.	Carry on with the lab. To start it again from the beginning, use Revert, then Start.
`No policy installation was started`	**New** (menus). Nothing changed, so nothing was installed.	Nothing to do.
`ERROR: ...` then `Setup ended`	Something failed. Nothing was changed.	Read the reason on the next line. Close SmartConsole or publish your changes, then try again, or ask the instructor.
`Note: lab3a has no revert script...`	**New.** Only labs 2B and 8A have a revert script; for the others Revert runs the stop script.	Run Start again to set the lab up afresh.
`Next (lab guide step 4A-4.2): start CPD again`	**New.** Lab 4A's stop script does not start CPD.	Run the `cpwd_admin start` command shown, on that gateway.
`labXa is already running. Do you want to restart it?`	Labs 2A, 3A and 4A run in the background until stopped.	Answer `n` to leave it running.
`Interrupted - discarding changes and logging out`	**New.** You pressed Ctrl-C.	Nothing is left locked. Start the lab again.

5. What is different from the original Check Point scripts

Same commands, same lab problems, same normal messages. The changes only show when something unusual happens:

Situation	Original scripts	Updated scripts (2.2.5)
Running `labstart` a second time (6A, 8A, 9A)	Prints only `Setup ended`, with no reason. 9A shows an error.	Explains that the lab is already set up and how to start it again
`labrevert` on a lab without a revert script	Silently runs the stop script	Says so, and tells you to run Start again
Ctrl-C during a lab script	Can leave objects locked in SmartConsole, or (Lab 2B) A-SMS-02 cut off	Discards the session and logs out; Lab 2B restores the route to A-SMS-02
A failure	Some failures print no reason, or report success	Always `ERROR:` plus the reason from the management server
Lab 2B prompts	Easy to type exit in the wrong window	One extra line says which window to type exit in
Lab 4A stop	Silent about CPD	Reminds you of step 4A-4.2
`-v` before the lab name	Only the launcher is verbose	Works before or after the lab name
Running `labstop lab9a` twice	Error on the second run	Completes normally

6. Lab notes: where the lab and the book can differ

Lab / step	What you may see	What to do
2B, step 2B-1.3	Page 160 shows A-SMS as **No Communication** instead of a conflict.	Go back to the **first** A-SMS window, the one showing `Type "exit" to restore connectivity`, and type `exit` there. Allow 3 to 5 minutes for the conflict to show.
2B, step 2B-2.32	View Sessions also lists an (Unnamed) **WEB_API** session from 127.0.0.1 with 0 changes.	A background system login. Ignore it or disconnect it; it closes by itself after about 10 minutes.
2B, step 2B-2.37	**Sync Peer** is greyed out; A-SMS shows *Successfully synced at ...*	The automatic sync has already run. Skip 2B-2.37 and 2B-2.37a.
2B, Task 2B-3	A-SMS says `Kindly run the stop script on A-SMS-02 first`	Both servers are still Active. Run the stop script on A-SMS-02, then on A-SMS again (a few minutes later if it repeats).
4A, Task 4A-4	After the lab, policy installs to A-GW-Cluster fail with *TCP connectivity failure (port 18191)*.	CPD was not started again (step 4A-4.2). Run the `cpwd_admin start` command on the gateway.
6A, stop	Recent Tasks shows *A-GW-01/02 - kernel parameters* stuck at 10%.	Harmless: the parameter is already reset. Clear the tasks.
8A, Tasks 8A-1/8A-2	Bravo-Standard fails on B-GW: *rule with Access Role includes targets that do not support Access Role*.	Intended. This is the problem Lab 8A asks you to troubleshoot.
Any lab	A policy install fails on A-GW-Cluster straight after another action.	The previous lab or installation was still running. Stop the previous lab first, and use a menu (they wait for installations).
Any lab	B-GW: *policy versions are incompatible ... Disable Accelerated Install Policy* (sk168055).	Happens after an interrupted installation. Do one full (non-accelerated) install of Bravo-Standard.

7. For instructors

Improved scripts and labmenu (A-SMS, A-SMS-02, A-GW-01, A-GW-02): run in PowerShell on A-GUI; about 20 seconds per host; the password is asked once and not stored:

```
irm https://raw.githubusercontent.com/Caminoltd/CCTE-R82-LabScripts/main/Install-CCTEScripts.ps1 | iex
```

CCTE Lab Menu on A-GUI (separate and optional; installs nothing on the appliances):

```
irm https://raw.githubusercontent.com/Caminoltd/CCTE-R82-LabScripts/main/agui/Install-CCTELabMenu.ps1 | iex
```

Undo: run either installer with `-Restore` (scripts) or `-Uninstall` (A-GUI menu); see the README on GitHub.
Check a host: `cat /home/admin/CCTE_Scripts/VERSION` shows `2.2.5`. Admin logs in to clish on the appliances, so type `expert` first. Verbose output and the last API reply are in `/tmp/labXa.json` and `.err`. If Lab 8A start reports a timeout querying `alpha.cp`, AD is still waking up after a lab resume: wait a minute and run it again. The ATC portal download of `lab8a_start` has a typo (TikTokd, line 74); the lab image and this set are correct.

Source code, change list and originals: github.com/Caminoltd/CCTE-R82-LabScripts.